

PENTECOST UNIVERSITY



SOWUTUOM, ACCRA, GHANA

FACULTY OF ENGINEERING, SCIENCE AND COMPUTING

DEPARTMENT OF INFORMATION TECHNOLOGY

TWI ALPHABET AND NUMERAL RECITER
(CASE STUDY OF JAMVERSON ACADEMY)

FINAL YEAR PROJECT REPORT SUBMITTED IN PARTIAL FULFILMENT
FOR THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF BSC.
INFORMATION TECHNOLOGY

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JUNE, 2026

DECLARATION

We hereby declare that this project work herein submitted as the Final-Year Project Report in partial fulfillment of the requirement for the award of Bachelor of Science degree in Information Technology is the result of our own research. Therefore, no part of this project work has been presented whole or in part to any other institution for any recommendation and that the project has been duly supervised and assessed by Mr. Joshua Nettey in accordance with the laid down guidelines of Pentecost University.

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CERTIFICATION

I hereby certify that the preparation and presentation of the project were supervised in accordance with the guidelines on supervision laid down by Pentecost University.

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DEDICATION

This project is first and foremost dedicated to Almighty God, whose grace, wisdom, and strength have carried us through every step of this journey. Without His guidance, protection, and favour, this achievement would not have been possible. We are deeply grateful for His unfailing love and the divine ability He granted us to overcome challenges and complete this work.

We also dedicate this project to our cherished families, whose unwavering love, sacrifices, and constant encouragement formed the backbone of our success. Your prayers, support, and belief in us gave us the courage to persevere even when the road became difficult. We are forever thankful for the values you instilled in us and the strength you helped us build.

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ABBREVIATIONS

FESAC – Faculty of Engineering Science and Computing.

EXECUTIVE SUMMARY

This project presents the design and development of a Twi Alphabet and Numeral Reciter System intended to enhance the teaching and learning of the Twi language at Jamverson Academy. Presently, Twi instruction relies predominantly on traditional pedagogical approaches such as chalkboard writing and oral repetition. Although this method is cost-effective and allows teachers to control lesson pacing and provide immediate feedback, it lacks technological support required for effective pronunciation mastery, learner engagement, and systematic performance assessment. A critical limitation of the existing approach is the absence of audio-visual reinforcement, which negatively affects pupils' ability to accurately pronounce Twi alphabets, words, and numerals without direct teacher intervention. Furthermore, learning activities are confined to the classroom environment, offering limited opportunities for independent practice. Assessment and progress tracking are also largely manual and oral, making it difficult to evaluate learner performance objectively and over time.

To address these challenges, the project aims to develop an interactive reciter system that integrates digital audio, structured content delivery, and learner progress monitoring. The system will implement automated recitation of Twi alphabets, phonetic spelling of two-to-five letter words, audio supported counting from 1 to 100, and numeral spelling from 1 to 50. Additionally, the system will include a performance tracking module that records pupil interaction data, enabling teachers to assess learning progress, identify areas of difficulty, and make data-driven instructional decisions.

By leveraging educational technology, the proposed system seeks to improve pronunciation accuracy, increase learner engagement through repeated audio exposure, and support self-paced learning beyond the classroom. The successful implementation of this system will modernize Twi language instruction, enhance instructional efficiency, and contribute to improved learning outcomes at the lower primary level.

CHAPTER ONE

INTRODUCTION

1.1 Background

Language learning is a key part of education for pupils in Primary 1 to Primary 3. At this level, pupils begin to develop basic skills in reading, speaking, spelling, and counting. Teaching methods used for these classes must be simple, engaging, and easy to follow. In recent times, digital learning tools have been introduced to support pupils by combining sound, text, and images. These tools help pupils hear correct pronunciation, see letters and numbers clearly, and practice repeatedly, which improves their understanding and retention [1].

For local languages such as Twi (Akan), digital learning resources are still limited, even though the language is widely spoken in Ghana. Many pupils at this level face difficulties in recognizing letters, pronouncing words correctly, and counting numbers confidently. These challenges can affect their ability to read and write effectively. Research have shown that children perform better when teaching methods include listening, repetition, and visual aids rather than relying only on traditional board work or rote learning [2]. However, such interactive tools are not always available in many basic schools.

In the African and Ghanaian context, several efforts have been made to develop language resources for local languages, including Twi. Most of these efforts focus on text translation or academic research, rather than tools designed specifically for young learners. As a result, there are few systems that help pupils practice reciting alphabets, spelling simple words, or counting numbers with correct pronunciation [3]. This gap makes it difficult for children to practice Twi effectively outside the classroom. In many Ghanaian basic schools, especially at Jamverson Academy, teachers have observed that pupils often forget alphabets, confuse letters, mispronounce words, and counting of number in Twi challenging. Spelling simple Twi words can also be challenging, and counting numbers beyond the early stages becomes difficult without repeated practice. Although teachers use methods such as songs, and oral repetition, these methods alone are not always enough to meet every learner's needs [2]. In addition, assessment of pupils' progress is often informal and does not fully capture pronunciation or continuous improvement.

Therefore, there is a clear need for a simple and interactive Twi Alphabet and Numerals Reciter designed specifically for lower primary pupils. Such a system can support learning by providing clear audio pronunciation, visual display of alphabets and numbers, spelling guidance, and basic performance tracking. This project aims to address the challenges identified by developing a child-friendly tool that supports early literacy and numeracy in Twi. By doing so, it will help improve pupils' confidence, strengthen their foundation in the Twi language, and support teachers in delivering more effective instruction.

1.2 Problem Statement

The teaching of the Twi language at Jamverson Academy, particularly among pupils in Classes 1–3, is predominantly based on traditional instructional methods such as board writing, oral repetition, and occasional use of songs and rhymes. While these approaches promote classroom participation, they are insufficient in ensuring effective learning outcomes for young learners. Evidence from the study indicates that pupils frequently forget previously learned alphabets, struggle with accurate pronunciation, and often confuse similar letters. The lack of interactive and audio-visual learning tools limits pupils' ability to practice independently and hinders retention. Consequently, the current system does not adequately support the cognitive and linguistic development required for early language acquisition.

In addition, pupils encounter significant difficulties in spelling and word recognition in Twi. Teachers rely heavily on writing words on the board and assessing pupils through oral repetition and individual questioning, yet many learners continue to mix up letters and incorrectly apply English phonetics to Twi words. The absence of structured and engaging spelling support tools further complicates the learning process. Moreover, there is no standardized or technology-supported approach to monitor individual progress, making it difficult for teachers to identify learning gaps and provide timely intervention. This places an additional burden on teachers while limiting pupils' ability to learn at their own pace and improve consistently.

Similarly, although pupils demonstrate basic understanding of Twi numerals within the range of 1–20, they face increasing challenges when progressing to higher numbers up to 100. Their dependence on written forms and lack of continuous reinforcement contribute to poor retention

and comprehension. Current assessment methods, including class exercises and examinations, are constrained by factors such as pupil shyness and limited participation, making performance tracking inefficient and incomplete. Therefore, there is a critical need for an innovative, interactive, and learner-centered solution that integrates audio-visual learning, supports pronunciation and spelling, enhances numeracy skills, and enables effective monitoring of pupil performance. This gap justifies the development of a Twi Alphabet and Numeral Reciter mobile application tailored to the needs of lower primary pupils at Jamverson Academy.

1.3 Objectives

The main objective is to construct an alphabet and numeral reciter. To achieve this, the developers will construct a system that will

- Recite the Twi alphabets
- Recite and spell words (Two to Five letters)
- Recite the Twi counting numbers (1 to 100)
- Recite and spell numerals (1 to 50)
- Access pupil's development progress as well as their performance.

1.4 Scope

The scope of this project covers the design and development of a Twi Alphabets and Numbers Reciter application aimed at enhancing early childhood learning of the Twi language. The project focuses on five core functions:

- Recites Twi alphabets, spelling and pronunciation of two-to-five letter words, recites counting numbers from 1 to 100, spells numerals from 1 to 80, and assesses children's learning progress.
- The system incorporates visual aids, audio pronunciation, and interactive activities to make learning engaging and effective.
- It is designed for lower primary pupils, teachers, and parents to support both classroom and home learning. The project also includes basic data recording features for tracking learners' performance and improvement over time.

1.5 Significance of the Project

This project is important because it provides an interactive and easy-to-use tool that helps pupils learn Twi alphabets, words, and numerals more effectively. By combining audio, visual, and interactive features, the system supports learning both in school and at home. The significance of the project includes the following:

- Parents or guardians can use the Twi Alphabets and Numbers Reciter at home to assist their children in improving their learning skills.
- Pupils can learn correct pronunciation of alphabets, words, and numerals independently, without constant help from teachers or parents.
- The system helps pupils recognize, memorize, and spell Twi alphabets and words more effectively through audio-visual support.
- It supports learning of Twi counting numbers from 1 to 100, strengthening early numeracy skills.

1.6 Organization of the Study

This project report is structured into five chapters. Chapter one introduces the study, presenting the background, problem statement, objectives, scope, limitations, and significance of the project. Chapter Two reviews related literature on Twi language learning, digital learning tools, and existing systems that support indigenous language education. Chapter Three outlines the research methodology, including data collection methods, system requirements, and design processes used in developing the Twi Alphabets and Numbers Reciter. Chapter Four focuses on the system development, detailing the design, implementation, testing, and evaluation of the application. Finally, Chapter Five presents the summary, key findings, conclusions, and recommendations for future improvements and further research in Twi language learning and educational technology.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

In order to further inform the design of the proposed Twi Alphabet and Numeral Reciter, additional Akan/Twi-based learning systems were reviewed. These systems include Kiddie Twi First Words, Speak Akan Language, Uncle Dan Kids, and Akan Twi Guide. The review examined their features, strengths, and limitations to identify gaps relevant to lower primary education, interactivity, and continuous recital.

2.2 Current System

At Jamverson Academy, Twi is mainly taught through traditional methods such as board writing and oral repetition, where teachers write Twi alphabets and numbers on the board for pupils to repeat after them. This approach is low-cost, simple, and encourages class participation. It also allows teachers to control the pace of lessons, provide immediate corrections, and maintain close interaction with pupils. However, the system has several weaknesses. Many pupils struggle with pronunciation without the teacher's assistance, and the lack of audio-visual support makes lessons less engaging. In addition, tracking pupil progress is difficult because assessments are mostly oral and manual, while learning remains limited to the classroom with few opportunities for self-practice.

2.3 Reviewed Systems

The existing systems reviewed were the Kiddie Twi First Words, Speak Akan Language, Uncle Dan Kids and the Akan Twi Guide. The details of the various systems including their features, strengths, and limitations are provided below.

2.3.1 Kiddie Twi First Words

Kiddie Twi First Words is an educational system developed to support language learning by introducing young learners to foundational Akan (Twi) vocabulary [4]. The system is specifically

designed for children, focusing on the acquisition of basic words through a combination of visual representations and audio pronunciation. By linking familiar objects, animals, and everyday items with their Twi equivalents, the platform enables children to develop both recognition and recall skills in a developmentally appropriate manner.

The system emphasizes multisensory learning, combining images and spoken words to reinforce memory retention and understanding, which is critical in early language education. Audio pronunciation ensures that learners are exposed to correct intonation and tonal patterns inherent in Twi, fostering accurate speech development from an early age. Furthermore, the interactive design encourages independent exploration, allowing children to engage with the system at their own pace, promoting self-directed learning and curiosity.

Overall, Kiddie Twi First Words provides a structured and engaging introduction to the Akan language, laying a strong foundation for further language acquisition while promoting cultural familiarity. The system is a valuable tool for early learners, combining educational effectiveness with age-appropriate accessibility [4].

Key Features of Kiddie Twi First Words

The key features of Kiddie Twi First Words include the following:

- It uses colorful images to represent common everyday objects.
- The system provides Twi audio pronunciations for each word.
- It has a simple and child-friendly interface.
- It supports independent learning with minimal adult assistance.
- It promotes basic vocabulary recognition through repetition.

Strengths of Kiddie Twi First Words

The strengths of Kiddie Twi First Words are as follows:

- It introduces children to indigenous language learning for pupil.
- The visual–audio association enhances memory and word recognition.
- The simple interface makes it easy for young learners to navigate.
- It supports foundational language exposure suitable for early learners.

Limitations of Kiddie Twi First Words

Despite its strengths, Kiddie Twi First Words has the following limitations:

- It does not support alphabet learning or continuous alphabet recital.
- The system does not include numeral learning or number recitation.
- It lacks assessment and progress-tracking functionality.
- There are no interactive exercises such as quizzes or tracing activities.
- The learning scope is limited mainly to vocabulary recognition.

2.3.2 Speak Akan Language

Speak Akan Language is an educational software system specifically developed to help children and non-native speakers learn the Akan (Twi) language in a practical and accessible way [5]. The system primarily targets users who have little or no prior knowledge of Akan and aims to provide foundational language skills for basic communication. By combining text, translation, and audio support, it enables learners to understand commonly used phrases, expressions, and everyday conversational language.

The system's content is structured to make learning gradual and manageable, allowing users to build confidence as they progress. It integrates audio pronunciation to guide learners in mastering correct intonation and pronunciation, which is particularly important in tonal languages like Twi. Additionally, Speak Akan Language serves as a self-paced learning tool, giving users the flexibility to study at their convenience without the need for a formal classroom environment. This approach is intended to foster both linguistic competence and cultural understanding, as learners are introduced not only to the language but also to common social expressions and communication patterns used in Akan-speaking communities.

Overall, Speak Akan Language provides a foundational learning experience for beginners, supporting both practical communication skills and exposure to Akan cultural contexts [5].

Key Features of the Speak Akan Language

The key features of Speak Akan Language include the following:

It provides commonly used Akan phrases with English translations.

- The system includes audio pronunciation to support correct intonation.
- It introduces basic conversational Akan expressions.
- Content is organized in a structured and accessible format.

- It serves as a self-study reference system.

Strengths of the Speak Akan Language

The strengths of Speak Akan Language include:

- It supports bilingual learning through English–Twi translation.
- Audio pronunciation aids understanding of tone and pronunciation.
- It is useful for basic communication and cultural exposure.
- The system is suitable for independent learners and beginners.

Limitations of the Speak Akan Language

The limitations of Speak Akan Language are as follows:

- It is not specifically designed for children or early learners.
- The interface lacks colorful visuals and animations suitable for kids.
- It does not support alphabet or numeral learning.
- There is no continuous alphabet or number recital feature.
- The system lacks interactive learning and assessment tools.

2.3.3 Uncle Dan Kids

Uncle Dan Kids is an educational media-based learning system designed to facilitate children acquisition of the Akan (Twi) language through engaging audio-visual content [6]. The system integrates songs, animations, and storytelling to teach foundational language concepts, including the alphabet, numbers, rhymes, and basic moral lessons. By combining entertainment with education, the platform creates an immersive learning environment that captures the attention of young learners while promoting comprehension and retention of language skills.

The system emphasizes multisensory learning, using visual stimuli alongside auditory cues to reinforce memory and understanding. Songs and rhymes provide repetitive and rhythmic exposure to Twi words, aiding in pronunciation, tone recognition, and vocabulary development. Storytelling elements introduce cultural context and ethical lessons, promoting both linguistic and moral education. Overall, Uncle Dan Kids offers a structured yet engaging approach to early Twi

learning, making it suitable for children, while supporting both language acquisition and cultural familiarity [6].

Key Features of Uncle Dan Kids

The key features of Uncle Dan Kids are as follows:

- The system uses animated visuals to engage learners.
- It promotes repetition and learning through music.
- It integrates cultural and moral education.
- It is accessible across multiple digital media platforms.

Strengths of Uncle Dan Kids

The strengths of Uncle Dan Kids include:

- Music and repetition enhance memory and recall.
- Audio-visual presentation sustains children's attention.
- It supports cultural identity and indigenous language preservation.
- It makes learning enjoyable and engaging for children.

Limitations of Uncle Dan Kids

Despite its strengths, Uncle Dan Kids has the following limitations:

- It lacks a structured and progressive learning path.
- The system does not provide interactive practice activities.
- Continuous alphabet or numeral recital is not controlled.
- Learning is largely passive and media-dependent.

2.3.4 Akan Twi Guide

Akan Twi Guide is a structured language reference system developed to support learners in acquiring Akan (Twi) vocabulary, expressions, and basic grammar rules [7]. Intended primarily for self-study, the system provides clear explanations of grammatical rules alongside relevant examples, enabling learners to develop both comprehension and communication skills.

The platform allows users to consult it as a reference tool to reinforce learning, improve accuracy, and clarify usage, supporting both beginners and more advanced learners. By presenting vocabulary and expressions commonly used in Akan-speaking communities, Akan Twi Guide promotes cultural familiarity while fostering linguistic competence. Its structured, self-paced approach makes it a reliable resource for independent learning, offering a foundation for continued language development in both practical and academic contexts [7].

Key Features of Akan Twi Guide

The key features of Akan Twi Guide include the following:

- It provides organized Akan vocabulary and expressions.
- The system includes basic pronunciation guides.
- It introduces simple grammatical explanations.
- It supports independent self-study.
- It serves as a language reference tool.

Strengths of Akan Twi Guide

The strengths of Akan Twi Guide are as follows:

- It provides structured foundational knowledge of Akan.
- The system is useful for reference and revision purposes.
- Content organization supports systematic learning.
- It aids learners interested in language structure.

Limitations of Akan Twi Guide

The limitations of Akan Twi Guide include:

- It is not suitable for class 1-3 learners.
- The system lacks audio-visual and animated learning aids.
- It does not support alphabet or numeral learning.
- There is no continuous recital functionality.
- The system lacks interactivity and assessment features.

2.4 Justification of the Proposed System

The review of existing Akan/Twi learning systems, including Kiddie Twi First Words, Speak Akan Language, Uncle Dan Kids, and Akan Twi Guide, revealed several functional gaps that limit their effectiveness for learning. Although these systems support basic vocabulary acquisition, cultural exposure, and pronunciation, they generally lack structured alphabet learning, continuous alphabet recital, and progressive numeral recitation. In addition, most of the reviewed systems do not provide learner assessment or progress-tracking features, making it difficult for teachers and parents to monitor learning outcomes. Some systems are also largely passive or text-based, which reduces interactivity and learner engagement for young children.

The proposed Twi Alphabet and Numeral Reciter is therefore justified as it directly addresses these identified limitations. The system provides continuous and uninterrupted Twi alphabet recital, progressive numeral learning, and a simple, colorful, and interactive interface designed specifically for early learners. It also incorporates assessment functionality to enable effective monitoring of learners' progress. By improving upon the strengths of the reviewed systems and resolving their weaknesses, the proposed system offers a more comprehensive, structured, and child-centered approach to Akan language learning.

2.5 Summary

The reviewed systems; Kiddie Twi First Words, Speak Akan Language, Uncle Dan Kids, and Akan Twi Guide show different approaches to supporting Twi language learning. Kiddie Twi First Words uses images and audio for vocabulary learning, while Speak Akan Language focuses on bilingual communication skills. Uncle Dan Kids uses songs, animations, and rhymes to teach children, whereas Akan Twi Guide serves mainly as a self-study reference tool. Although these systems are useful for language exposure and cultural promotion, most lack structured alphabet lessons, numeral learning progression, learner assessment, and progress tracking. Their limited interactivity also reduces effectiveness for young learners. These gaps justify the need for a more structured and child-centered Twi learning system.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

The development of the Twi Alphabet and Numeral Reciter application follows the Scrum Framework, an iterative and flexible software development approach. Scrum is adopted to support continuous stakeholder involvement, especially teachers and pupils, while ensuring incremental delivery of system features. This approach allows the development team to quickly respond to feedback and improve system quality [8]. Through short cycles called sprints, features such as alphabet recitation, word spelling, numeral pronunciation, and performance tracking are gradually developed, tested, and refined. Regular reviews ensure usability and accuracy. Overall, Scrum helps deliver an effective, interactive learning tool that supports Twi literacy, numeracy, and improved classroom learning outcomes.

3.2 Justification for Choosing the Scrum Framework

The Scrum Agile Framework was selected for the Twi Alphabet and Numeral Reciter project due to its advantages over traditional linear methodologies such as the Waterfall model. These advantages are particularly relevant given the evolving requirements, user diversity, and educational focus of the project.

Firstly, Scrum supports adaptive planning and flexibility. In educational environments, especially those involving young learners, system requirements may change as pupils interact with early versions of the application. Scrum accommodates such changes through short, time-boxed sprints that allow the development team to incorporate feedback without disrupting the overall project schedule [8]. This flexibility ensures that the final system remains relevant, learner-centered, and effective in addressing pronunciation and numeracy challenges.

Secondly, Scrum emphasizes the incremental delivery of value. Instead of waiting until the entire system is completed, functional modules such as the Alphabet Reciter or Number Reciter are developed and tested early in the project. This enables early demonstrations, feedback collection, and validation of features like pronunciation accuracy, spelling logic, and user engagement [9].

Incremental delivery reduces development risks and ensures that errors are identified and corrected at early stages.

Thirdly, Scrum promotes collaboration and risk mitigation. Teachers and pupils are involved from the Initiation phase through Sprint Reviews, ensuring that the system meets real classroom needs. Teachers contribute linguistic expertise by validating Twi pronunciations and spellings, while pupils provide feedback on usability and engagement. This collaborative approach minimizes the risk of developing features that are technically sound but educationally ineffective [8].

Finally, Scrum enhances quality assurance and transparency through frequent reviews, retrospectives, and testing cycles. Progress is continuously monitored, and development practices are improved after each sprint. The adoption of Scrum is therefore a strategic decision aimed at delivering a high-quality, user-validated educational application that supports effective teaching, learning, and assessment [9].

3.3 Comprehensive Scrum Methodology Activities for the Twi Alphabet and Numeral Reciter Project

The development of the Twi Alphabet and Numeral Reciter application follows the Scrum Agile Framework throughout its lifecycle. Scrum is widely recognized for managing complex software projects through iterative development, continuous feedback, and clearly defined roles and artifacts [8]. This methodology ensures that all project objectives including alphabet recitation, word spelling, numeral recitation, and pupil performance tracking are achieved systematically while allowing flexibility for improvement based on classroom testing and user input [9].

The primary objective of the project is to develop an interactive software application that enables pupils to recite and spell Twi alphabets and numerals accurately. Supporting objectives include:

- Reciting the Twi alphabets accurately.
- Reciting and spelling simple Twi words consisting of two to five letters.
- Reciting Twi counting numbers from 1 to 100.
- Spelling Twi numerals from 1 to 50.
- Monitoring and assessing pupil performance and learning progress.

To achieve these objectives, the project is structured into five major Scrum phases: Initiation, Sprint Planning, Execution, Sprint Review and Retrospective, and Release [8]. Each phase has clearly defined activities, responsibilities, and deliverables to ensure alignment with both educational goals and technical requirements [9].

3.3.1 Initiation Phase

The Initiation Phase establishes the foundation of the project by defining the problem statement, confirming the project scope, identifying stakeholders, and outlining the resources required for successful implementation [8]. This phase ensures that all participants have a shared understanding of the project's purpose, scope, and constraints.

A. Problem Definition and Scope Confirmation

Problem identification begins with an analysis of the learning challenges faced by pupils at Jamverson Academy. Traditional Twi language teaching methods, which rely heavily on board writing and oral repetition, often result in inconsistent pronunciation, limited engagement, and inadequate assessment of individual pupil progress [9]. These limitations create a need for an interactive and standardized digital learning tool.

The project addresses these challenges by developing a Twi Alphabet and Numeral Reciter application that includes:

- Interactive modules for alphabet recitation, word spelling, and numeral counting.
- Automated tools for recording and monitoring pupil progress.
- A child-friendly interface designed with strong visual and auditory appeal.

Stakeholder roles are clearly defined:

- Developers are responsible for system design, module implementation, integration, and testing.
- Supervisor provides academic oversight and methodological guidance.
- Teachers validate Twi pronunciations and evaluate educational content.
- Pupils act as end-users and provide usability and engagement feedback.

Scope confirmation involves documenting project boundaries, specifying supported alphabets, words, and numerals, and detailing expected functionalities such as recitation playback, spelling validation, and performance tracking.

3.3.2 Requirements Gathering and Analysis

Requirements gathering employs a mixed-method approach involving interviews, classroom observation, and document analysis to capture both functional and non-functional requirements.

3.3.2.1 Result from the interview

3.3.2.1.1 Reciting Twi Alphabets

Participant 1:

I write on the board, pronounce the words, and have pupils repeat after me. However, I've noticed that they tend to forget quickly, so I make it a point to revise with them for about 10 minutes. They really respond well when we incorporate fun activities like singing rhymes and picture reading - for example, showing them a picture of a mango and saying "this is a mango" brings a lot of joy to the learning process [10].

Participant 2:

When teaching Twi, I pronounce and point to each alphabet on the board, having pupils repeat after me. However, they often struggle to memorize and pronounce the alphabets correctly, frequently mixing them up. To address this, I rely heavily on writing on the board, which has proven to be an effective tool in supporting their learning. Notably, pupils enthusiastically engage

with activities involving reciting or singing the alphabets. For future improvements, it would be beneficial for the system to distinguish between capital and small letters, enhancing pupils' understanding of Twi orthography [10].

Participant 3:

The school supports Twi language teaching in lower primary by providing necessary textbooks. However, pupils sometimes struggle to recite the Twi alphabets, often mixing them up. Currently, the school relies on Twi textbooks as a resource. A Twi reciter could fill gaps by aiding pronunciation and spelling. This tool would fit well into the school's language curriculum, particularly when teaching Twi alphabets and numerals, making learning easier and more engaging [10].

3.3.2.1.2 Reciting and spelling words

Participant 1:

I write the keywords on the board and break them down for pupils to pronounce, but they often struggle with spelling, especially when words are long or when they apply English sounds to Twi words. To assess their spelling ability, I call them out individually, but unfortunately, their skills are lacking. I believe prioritizing simple words would be beneficial. Incorporating pictures and poems into a Twi reciter system would make spelling activities more engaging and fun for pupils [10].

Participant 2:

I introduce new Twi words by writing them on the board and having pupils repeat after me. However, they often struggle with spelling due to mixing up letters. To assess their spelling, I use techniques like having them spell words with their backs turned or after erasing the words from the board. I believe prioritizing simple and familiar words would be beneficial, and adding pictures to a Twi reciter would make learning fun [10].

Participant 3:

From the school's perspective, spelling in Twi is crucial as it's an integral part of the culture. However, teachers face challenges due to a lack of guiding materials. Currently, there are no

school-wide approaches or materials for spelling. A Twi reciter could support efforts by making spelling easy and engaging. For this age group, simple and familiar words that are easy to pronounce and spell would be most suitable [10].

3.3.2.1.3 Reciting counting numbers and numerals

Participant 1:

I write the number and spell it in Twi. Learning Twi numerals is easy for pupils. I reinforce learning through songs and games. A reciter system should support learning numerals from 1 to 100, presenting the number and its name [10].

Participant 2:

I write numerals on the board, point to them, and have pupils repeat after me. Pupils grasp Twi numerals from 1-20 but struggle beyond that and have trouble distinguishing numbers without written form. I don't reinforce learning outside the classroom. A reciter system supporting Twi numerals from 1-100 would be beneficial, presenting numerals in English, translating to Twi, and including spelling [10].

Participant 3:

Twi numeracy is integrated into early learning through the teacher's lesson notes. However, pupils struggle with Twi numerals from 20-100. Teachers currently find their own ways to teach numerals effectively. A Twi reciter could strengthen the numeracy program by making learning easy and improving pupils' skills to different standards. For numeral recitations, it's essential to have accurate audio with correct pronunciation [10].

3.3.2.1.4 Assessing development and performance

Participant 1:

I assess pupils' progress by calling them to read, and adding songs to lessons helps track their progress. However, they struggle to speak up due to shyness. A Twi reciter could aid assessment by incorporating pictures and songs, making learning engaging. Useful features would include pronunciation practice, such as fruits and vegetables [10].

Participant 2:

I assess pupils through examinations and class exercises, and track progress by giving home assignments and using them as class exercises. Challenges arise when pupils struggle with exercises despite previous lessons. A Twi reciter would ease workload and enable pupils to learn independently, building fluency. Useful features would include breaking words into syllables [10].

Participant 3:

The school monitors pupils' progress in Twi language and numeracy through examinations and oral exercises. Common assessment methods include class exercises and home assignments. However, current assessment practices face challenges, making it somewhat difficult. A Twi reciter could be integrated into assessment processes by incorporating audio and interactive elements [10].

Functional Requirements

- Accurate audio recitation of Twi alphabets.
- Spelling of simple Twi words (2–5 letters).
- Recitation of numbers from 1 to 100.
- Spelling of numerals from 1 to 50.

Non-Functional Requirements

- High-quality audio output for clear pronunciation.
- Fast system response suitable for classroom use.
- Compatibility with commonly used school devices.

3.3.3 Scrum Artifacts Creation

Scrum artifacts are created to ensure transparency, planning accuracy, and progress tracking. The Product Backlog is developed using User Stories, which describe system features from the user's perspective [11]. Examples include:

“As a pupil, I want to recite alphabet ‘A’ so that I can learn its correct pronunciation.”

“As a pupil, I want to spell the word ‘Adee’ to improve my reading skills.”

“As a teacher, I want to record and view pupil scores to track learning progress.”

User Stories are prioritized based on educational importance and development feasibility, guiding sprint planning and execution [11].

3.3.4 Sprint Planning Phase

The Sprint Planning Phase translates prioritized backlog items into actionable development tasks. This phase focuses on system architecture, interface design, and feature prioritization [8].

A. Design and Architecture Planning

System architecture planning defines the technical structure of the application and includes:

- Selection of appropriate development tools and platforms for multimedia learning.
- Design of database schemas for storing Twi alphabets, words, numerals, and performance metrics.
- Creation of wireframes and UI/UX mockups suitable for early learners.

B. Core Feature Planning

Each core module is planned in detail:

- Alphabet Module: visual-audio mapping of Twi characters.
- Numbers Module: recitation and numeral spelling logic.
- Progress Tracking Module: structures for storing scores, errors, and engagement data.

3.3.5 Execution Phase (Development Sprints)

The Execution Phase involves iterative development through multiple sprints, each lasting approximately two to three weeks. Daily Scrum meetings are held to monitor progress, address challenges, and ensure alignment with sprint objectives [8].

Sprint activities include:

- Development of the Alphabet Recitation Module.
- Implementation of the Numbers Recitation and Progress Tracking Module.
- Development of the Word Spelling Module.
- Implementation of the Numeral Spelling Module and system integration.

3.3.6 Sprint Review and Retrospective

At the end of each sprint, a Sprint Review is conducted to demonstrate completed features and gather stakeholder feedback. Based on this feedback, the Product Backlog is refined and reprioritized [8].

A Sprint Retrospective is then held to evaluate team performance, identify challenges, and define improvements for subsequent sprints, supporting continuous process improvement [9].

3.3.7 Release Phase

The Release Phase focuses on preparing the application for operational use.

A. Final Testing and Quality Assurance

- User Acceptance Testing with teachers and pupils.
- Validation of assessment and reporting features.
- Performance and reliability testing to ensure classroom readiness.

B. Deployment and Training

- Deployment of the application on target devices at Jamverson Academy.
- Training sessions for teachers and pupils on effective system use.

3.4 Summary

The Twi Alphabet and Numeral Reciter application is developed using the Scrum Agile Framework, an iterative approach that supports continuous improvement, collaboration, and gradual system delivery. It is suitable for educational systems because it adapts to changing learning needs and user feedback. The method allows modules such as alphabet recitation, spelling, numeral learning, and progress tracking to be developed and tested separately. Teachers help validate content accuracy, while pupils provide usability feedback to improve design and engagement. The process follows five phases: Initiation, Sprint Planning, Execution, Sprint

Review and Retrospective, and Release. This approach ensures the final system is reliable, effective, and suitable for classroom use at Jamverson Academy.

CHAPTER FOUR

SYSTEM ANALYSIS AND DESIGN

Should be between Chapter 4: 2000 and 2500 words

4.1 Introduction

Summary

CHAPTER FIVE

TESTING AND IMPLEMENTATION

Should be between Chapter 5: 1500 and 2000 words

5.1 Introduction

Testing

Unit Testing

Integrated Testing

Implementation

Training

Maintenance

Conclusion

Limitations

Recommendations

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APPENDIX

Programming Codes

Welcome (Splash) Screen Codes